

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1710

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Time Duration: 1 Hour
Total Marks: 30

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

*I SK Arshad Basha (192312336) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

SK. Arshad Basha

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RUBRICS FOR CALCULATION

Numerical Problems						
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding, misses key elements	Misinterprets or unable to understand problem	20
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	18
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	23
Accuracy of Calculation	20%	All calculations accurate, correct final answer	Minor calculation errors	Frequent errors, incorrect final answer	No valid calculations	18
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	14

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Question - 1 $\rightarrow$ Flight Delay Notification

If a flight is delayed then passengers are notified.

Flight 202 is delayed.

Let:-

* D: Flight 202 is delayed

* N: Passengers are notified

Then:-

• $D \rightarrow N$

• D

CNF

The Implication is converted as

$$D \rightarrow N \Rightarrow D \times N$$

Negated the goal

Goal: $\neg N$

Negated goal: $\neg \neg N$

clauses

• $C_1: \neg D \times N$

• $C_2: D$

• $C_3: \neg N$

Resolution

Resolve N with $\neg N$

$$N, \neg N \Rightarrow D$$

conclusion

Passengers are notified

Question 2 $\rightarrow$ Employee certification

① If an employee completes the training the employee receives a certification.

② Kevin completed the training.

Let:-

* $T \rightarrow$ Kevin completed the training

* $C \rightarrow$ Kevin receives certification

Propositional Logic

$$T \rightarrow C$$

T

CNF

$$T \rightarrow C \Rightarrow \neg T \vee C$$

Clause:-

$$C_1 = \neg T \vee C$$

$$C_2 = T$$

Negated the goal

goal

C

$\neg C$

Resolve:-

$$(\neg T \vee C), T \Rightarrow C$$

$$\neg T \vee C$$

Then

$$C \Rightarrow \neg T$$

Conclusion:-

Kevin receives a certification.

ion 3 $\rightarrow$ medical Attention Alert

If a patient has a fever, then the patient needs medical attention.

② Ravi has a fever

Goal

prove that Ravi needs medical attention

Propositional Representation

Let

F = Ravi has a fever

m = Ravi needs medical attention

Goal: m

$F \rightarrow m$

CNF Conversion

$F \rightarrow m$

$\neg F \vee m$

Clauses

$C_1 = \neg F \vee m$

$C_2 = F$

Negate the goal

m

$\neg m$

$\therefore C_3 = \neg m$

Resolution

$(\neg F \vee m), F$

$\therefore m$

$$c_4 = m$$

$$m_1 = m$$

conclusion

The Empty clause proves the contradiction

Hence

Ravi needs medical attention

Question 4 $\rightarrow$ library late fine

- ① If a book is returned late, the fine is charged.
- ② The book was returned late

Goal

prove that a fine is charged

propositional Representation

Let

* L = Book was return late

* F = Fine is charged

Statements

$$L \rightarrow F$$

L

Goal

F

CNF Conversion

$$L \rightarrow F$$

$$\therefore \neg L \vee F$$

$$C_1 = \neg L \vee F$$

$$C_2 = L$$

Negate the goal

goal

F

negated goal = $\neg F$

Resolution steps

$$C_3 = \neg F$$

Resolve:-

$$(\neg L \vee F), L$$

$\therefore F$

$$C_4 = F$$

$$F, \neg F$$

conclusion

Since the empty clause is obtained as a fine is charged

Hence

Thus, the goal is successfully proved.

Question 5 $\rightarrow$ Exam Hall Ticket Eligibility

① If a student completes all assignments then the student is eligible for exam

② If a student is eligible for an exam, then the student receives a hall ticket.

③ The student completed all assignments

Goal

To prove that student receives a hall ticket

Propositional logic

Let:

$A \rightarrow$ student completed all assignments

$E \rightarrow$ student is eligible for an exam

$H \rightarrow$ student receives a hall ticket

$A \rightarrow E$

$E \rightarrow H$

A

Goal

H

Convert Implication int CNF

$A \rightarrow E$

$P \rightarrow Q \Rightarrow \neg P \vee Q$

$\neg A \vee E$

$G = \neg A \vee E$

$E \rightarrow H$

$G = \neg E \vee H$

the goal

aled goal

$$1 = \neg H$$

= complete set of clauses is

$$= \neg A \vee E$$

$$= \neg E \vee H$$

$$3 = A$$

$$= \neg H$$

ply Resolution

$$= (\neg A \vee E)$$

$$3 = A$$

$$\therefore (\neg A \vee E), A \Rightarrow E$$

$$C5 = E$$

$$2 = (\neg E \vee H)$$

$$\therefore (\neg E \vee H), E \Rightarrow H$$

Conclusion

The Empty clauses the student receives a hall ticket

Hence

The successfully proved using Propositional Resolution